

Transfer condition

Same governed problem family (evidence -> claim), but a longer unfamiliar situation. No method label or first-move cue is shown on the attempt page.

ATTEMPT FIRST - NCERT Exemplar Unit 2 Q35

Iron filings and sulphur were mixed together and divided into two parts, A and B. Part A was heated strongly while Part B was not heated. Dilute hydrochloric acid was added to both parts and evolution of gas was seen in both cases. How will you identify the gases evolved?

Plan and solve without being told which representation to use:

What the learner had to select

- Treat part A and part B separately.
- Choose observation/test -> inference as the useful representation.
- Tie each gas identity to the correct chemical evidence instead of guessing from the surface story.

Part A - heated

Heating forms iron sulphide. With dilute HCl, the evolved gas is hydrogen sulphide.

Identification evidence:

- Characteristic rotten-egg / foul odour.
- Lead acetate test turns black (marking point in source answer).

Part B - unheated

The mixture still contains iron. Iron reacts with dilute HCl and evolves hydrogen.

Identification evidence:

- Hydrogen burns with a pop sound.

Robust-transfer check

The transfer is successful only if the learner selected the evidence -> inference route without a family label and kept A and B separate. The chemistry is not harder because of bigger numbers; the challenge is model selection and evidence discrimination.